

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination – 2023****Course: B. Tech.****Semester : VI****Branch : Electronics and Telecommunication Engg./ Electronics and Communication Engineering****Subject Code & Name: BTETPE603A Microprocessors & Microcontrollers****Max Marks: 60****Date: 20/01/2024****Duration: 3 Hr.****Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) State addressing modes of following instructions along with its size in bytes. i. MVI A, 55H ii. LDAX B iii. LXI H, 2000H	CO1	6
B) Draw pin diagram of an 8085 microprocessor and describe functions of pins HOLD, HLDA, CLK, ALE.	CO2	6
C) Draw timing diagram of an STAX B instruction.	CO1	6
Q.2 Solve Any Two of the following.		12
A) Assume that five 8-bit numbers are stored from 1000H to 1004H. Write an assembly language program to clear lower nibble of each number.	CO1	6
B) With suitable example explain PUSH and POP instructions of 8085 microprocessor.	CO2	6
C) Describe MOV A, M and LXI H, 2000H instructions of 8085 with suitable examples.	CO3	6
Q. 3 Solve Any Two of the following.		12
A) With neat diagram explain basic block diagram of an 8051 microcontroller.	CO4	6
B) With suitable examples of each describe any three addressing modes of 8051 microcontroller.	CO3	6
C) Write an 8051 assembly language program to transfer contents of Register Bank 0 to Register Bank 1. (Use only direct addressing mode instructions)	CO5	6
Q.4 Solve Any Two of the following.		12
A) Describe TMOD register of an 8051 microcontroller.	CO2	6
B) Describe SCON register of an 8051 microcontroller.	CO4	6
C) Write an 8051 C program to toggle all bits of P1 continuously every 500 ms. Use Timer 1, mode 1 to create the delay. Assume that XTAL = 11.0592 MHz	CO5	6
Q. 5 Solve Any Two of the following.		12
A) Write an 8051 C program to send letters 'X', 'Y', and 'Z' to the 16x2 LCD.	CO7	6
B) Explain BSR and Input-Output modes of 8255 with neat diagrams.	CO6	6
C) Interface a 7-segment display to (draw interfacing diagram) 8051 microcontroller and write a C program to display numbers from 1 to 3 continuously on it.	CO7	6

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